Supporting Information for Publication

High-throughput native peptide-contact pharmacophore scoring at library scale: a staged cascade and its efficiency-retention trade-off

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This document contains supplementary figures and tables that accompany the manuscript. Figures are numbered Figure S1 through Figure S5; tables are numbered Table S1 through Table S14.

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Supplementary Figures

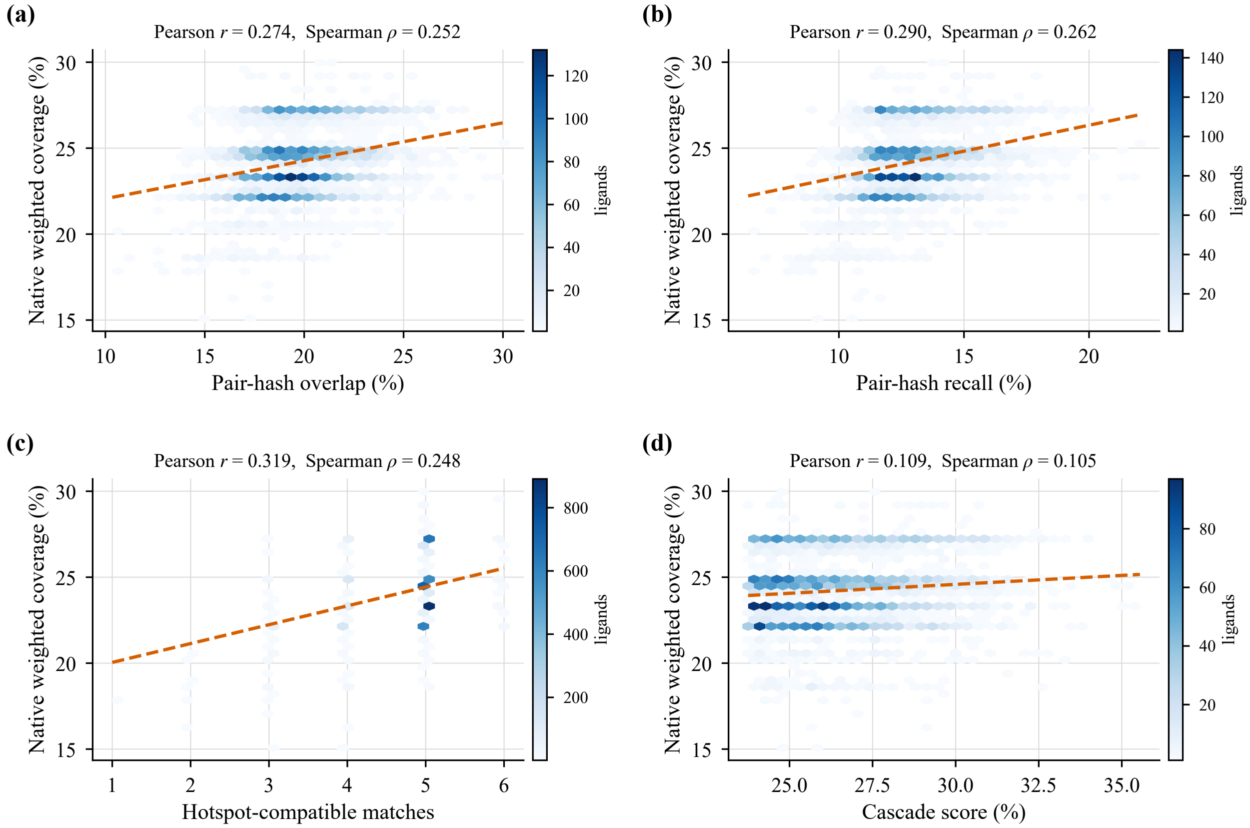


Figure S1: Upstream screen metrics against terminal native coverage across 4,997 successfully native-scored ligands. Hexbins show point density, dashed lines are least-squares fits, and each panel reports Pearson and Spearman correlations computed from the displayed cohort. These are internal associations among scored candidates and do not measure enrichment relative to molecules excluded upstream.

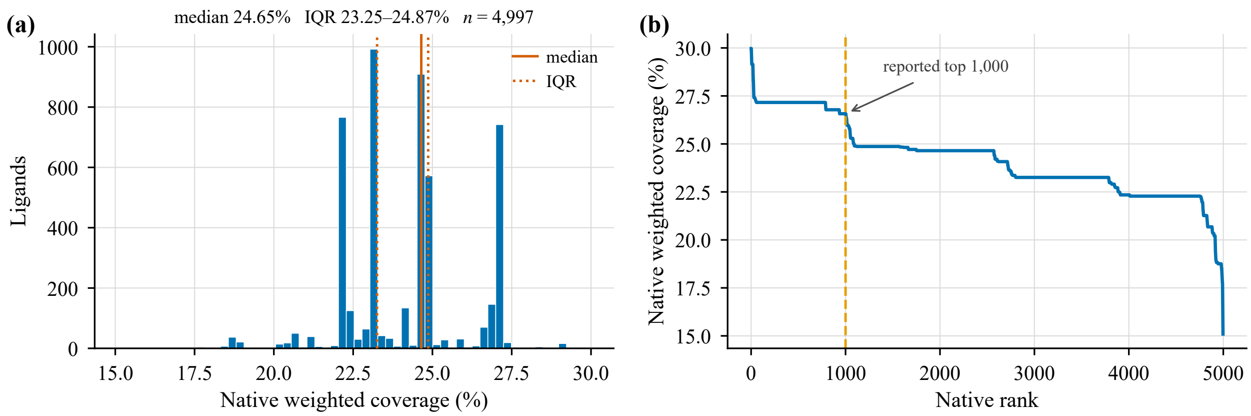


Figure S2: Distribution of native weighted coverage across 4,997 successfully scored ligands. (a) Histogram: median 24.65%, interquartile range 23.25–24.87%, and full range 15.09–29.97%. (b) Coverage in final-rank order, with the reported top 1,000 indicated.

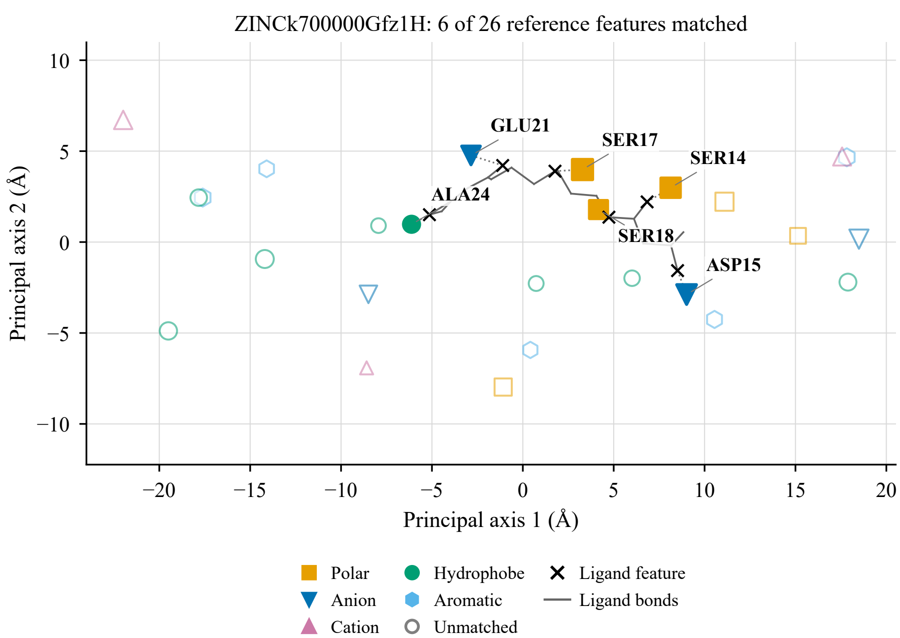
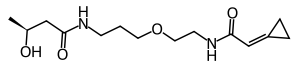
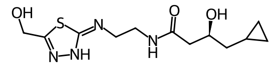


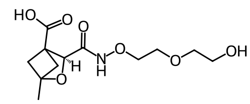
Figure S3: Best-scoring native overlay for the top-ranked ligand, ZINCk700000Gfz1H, against the retained GLP-1 peptide-contact reference from PDB 6X18. Filled markers denote matched reference features and open markers denote unmatched features. Shape and color indicate family; marker area indicates weight. Its 6 matched features give 29.97% coverage, native-fit RMSD 2.00 Å, and mean pair-distance error 1.69 Å. Gray bonds show the actual aligned heavy-atom ligand structure; black crosses mark its matched feature centroids, with dotted segments to the corresponding reference features. Coordinates are projected onto the first two principal axes of the reference. The reported fit metrics are three-dimensional; the aligned SDF independently reproduces the RMSD within its coordinate-rounding precision.



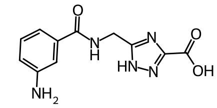
1. ZINCk700000Gfz1H
29.97% native coverage



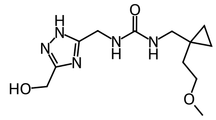
2. ZINCk700001bQgxo
29.97% native coverage



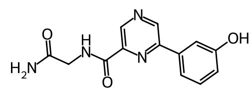
3. ZINCk500000BlHKL
29.53% native coverage



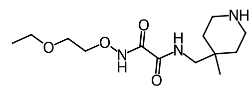
4. ZINCj600000JWMq0
29.15% native coverage



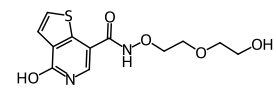
5. ZINCk500000gdMQW
29.15% native coverage



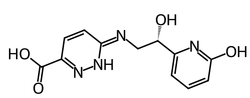
6. ZINCk600000EmAGZ
29.15% native coverage



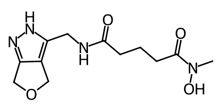
7. ZINCk5000003XjkE
29.15% native coverage



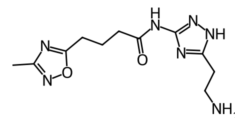
8. ZINCk800000gvPBH
29.15% native coverage



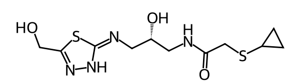
9. ZINCk600000Ja5bY
29.15% native coverage



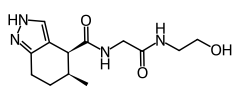
10. ZINCk600000y8tEo
29.15% native coverage



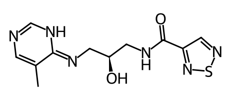
11. ZINCk50000045enc
29.15% native coverage



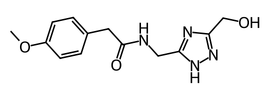
12. ZINCk5000009Gkrk
29.15% native coverage



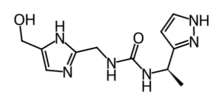
13. ZINCk500000448Ne
29.15% native coverage



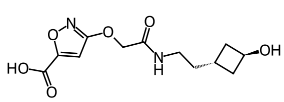
14. ZINCk5000009FEHo
29.15% native coverage



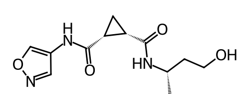
15. ZINCk7000004flr3
29.15% native coverage



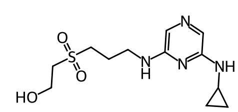
16. ZINCj700000JVICU
29.15% native coverage



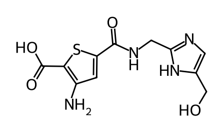
17. ZINCk6000018TA29
29.15% native coverage



18. ZINCj700000Emmu8
29.15% native coverage



19. ZINCk800000JUyax
29.09% native coverage



20. ZINCk700000JU4AJ
28.39% native coverage

Figure S4: Structures of the top 20 ligands ranked by native peptide-interface mimicry, in final-rank order from the million-compound screen. Each entry identifies the ligand and its native weighted coverage. Structural-alert matches remain annotations; the displayed ranks do not establish potency or assay suitability.

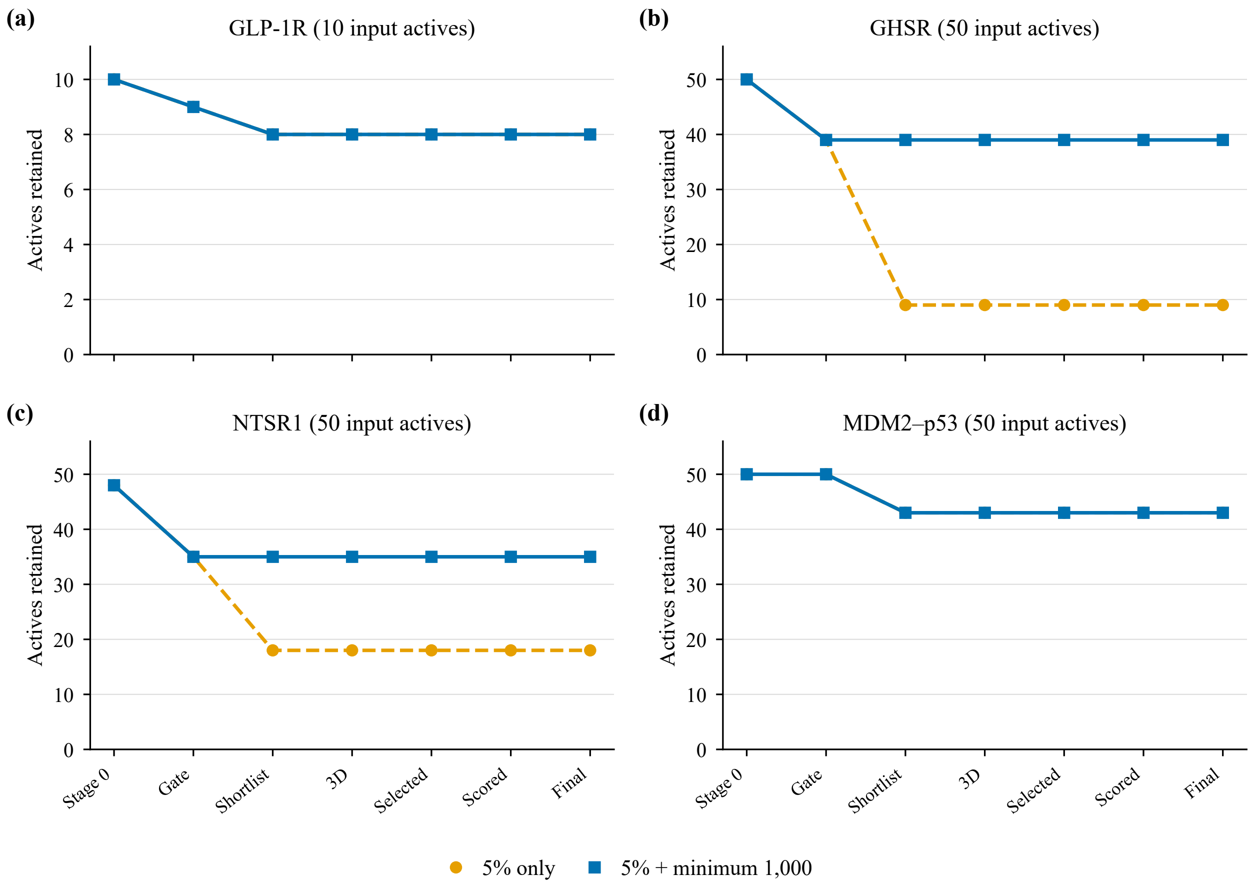


Figure S5: Active survival under each pair of shortlist rules. Both rules record structural alerts without exclusion and retain the property limits. Lines trace measured survival through gating, shortlisting, 3D scoring, native selection, successful native scoring, and the final ranking. The 1,000-molecule minimum is capped by the available candidates. The two curves coincide for GLP-1R and MDM2–p53.

Supplementary Tables

Table S1: Audited run configuration, counts, and timings for the GLP-1R screen. Stage-0–2 task times sum per-molecule elapsed times across workers and are not additive wall-clock phases. End-to-end wall time includes the native branch once; concurrent benchmark jobs share the workstation. No exhaustive million-ligand native-only runtime was measured.

Field	Value
Starting library	1,000,000 ZINC molecules; H17–H20
Workers	12 prescreen / 12 Stage 3 / 12 native
Stage-1 / Stage-2 query features	25 / 24
Shortlist rule	$\min(N_{\text{eligible}}, \max(\lceil 0.05 N_{\text{Stage0}} \rceil, 1000))$
Hotspot / pair weight	0.4 / 0.6
Typed-feature caps	Fixed: 2, 2, 4, 6, 6, 6
Stage-3 conformers / query anchors	16 / 28
Stage-3 distance / pair tolerance	2.75 / 2.75 Å
Pair-hash mode	precision_5bin
Structural alerts	Recorded without exclusion; native PAINS filter disabled
Native pool	12,000 Stage-3; 4,000 hotspot breadth; 4,000 native-supported hotspot
Native selection	Up to 5,000 ligands; maximum eight per Murcko scaffold
Native preparation	Up to two charge states; eight tautomers per charge state; four conformers per state
Final rank mode	native_first
Stage-0 pass	997,590
Stage-1 pass	990,191
Shortlist	49,880
Successful Stage-3 ligands	49,757
Native candidate pool	20,000
Selected native ligands	5,000
Prepared states	43,545
Scored states	43,473
Successfully native-scored ligands	4,997
Final ranking	1,000
Stage-0 accumulated task time	10,121.6 s
Stage-1 accumulated task time	998.5 s
Stage-2 accumulated task time	407.3 s
Stage-3 wall time	7,875.3 s
Native selection wall time	6.1 s
Native preparation wall time	652.0 s
Native scoring wall time	2,617.9 s
Native branch total wall time	3,300.0 s
End-to-end wall time	**12,336.7 s (3.43 h)**

Table S2: Curated GLP-1R contact groups and highest-weight receptor-side pharmacophore features. The listed entries illustrate the receptor-side hotspots that seed the cascade and anchor the curated contact logic used in Stage-1 gating and Stage-2 pair selection. Weights correspond to the receptor-side pharmacophore model used in the analyzed run, and the final row summarizes the curated residue groups in that model. Stages 1 and 2 do not select these features by weight: features are first collapsed to one per residue and complementary family, then ordered by curated priority (see Methods).

Feature type	Receptor atom	Weight	Curated group	GLP-1 residues represented
negative	GLU128 OE1	6.35	ECD anchoring	Phe28, Ile29, Leu32, Val33
positive	ARG299 NH1	5.91	Upper TMD activation pocket	His7, Glu9, Thr13, Ser14, Ser17, Ser18
negative	ASP67 OD1	5.85	ECD anchoring	Phe28, Ile29, Leu32, Val33
positive	ARG121 NH1	5.84	ECD anchoring	Phe28, Ile29, Leu32, Val33
positive	LYS197 NZ	5.76	Upper TMD activation pocket	His7, Glu9, Thr13, Ser14, Ser17, Ser18
positive	ARG190 NH1	5.63	Upper TMD activation pocket	His7, Glu9, Thr13, Ser14, Ser17, Ser18
aromatic	HIS212 CD2	5.29	ECL1 support	Trp31
aromatic	TYR145 CD1	5.23	Upper TMD activation pocket	His7, Glu9, Thr13, Ser14, Ser17, Ser18
aromatic	TYR145 CZ	5.23	Upper TMD activation pocket	His7, Glu9, Thr13, Ser14, Ser17, Ser18
aromatic	TYR69 CD1	5.13	ECD anchoring	Phe28, Ile29, Leu32, Val33
aromatic	TYR69 CZ	5.13	ECD anchoring	Phe28, Ile29, Leu32, Val33
aromatic	TRP39 CD1	5.12	ECD anchoring	Phe28, Ile29, Leu32, Val33

Curated contact groups represented in the receptor model: ECD anchoring = LEU32, TRP39, ASP67, TYR69, ARG121, LEU123, and GLU128; Upper TMD activation pocket = TYR145, ARG190, LYS197, TRP297, THR298, ARG299, LEU388, and SER392; ECL1 support = GLN211 and HIS212.

Table S3: Ligands showing the largest rank changes under native reranking. The first five rows show the largest upward shifts and the next five the largest downward shifts. Both ranks are defined over the same successfully native-scored cohort; positive shifts mean improvement.

Ligand ID	Screen rank	Native rank	Shift	Screen (%)	Native (%)
ZINCj700000JVICU	4,955	16	+4,939	12.08	29.15
ZINCk800000Pu2Q9	4,947	47	+4,900	12.09	27.23
ZINCk700001bP5R6	4,961	188	+4,773	12.07	27.16
ZINCk5000009FOpe	4,895	133	+4,762	12.15	27.16
ZINCk700000PBZyr	4,771	26	+4,745	12.28	27.81
ZINCk8000004kfyb	98	4,977	-4,879	14.81	18.75
ZINCk6000004dK2H	111	4,954	-4,843	14.77	18.75
ZINCj800000y9aPr	14	4,789	-4,775	15.43	21.27
ZINCk500000dpwFL	110	4,848	-4,738	14.78	20.68
ZINCk700000wnAUg	176	4,902	-4,726	14.59	20.21

Retrospective Benchmark Construction

The GLP-1R benchmark was assembled from ChEMBL target CHEMBL1784 and the local 1,000,000-molecule ZINC screening pool. Activity rows were retained only when they represented direct small-molecule GLP-1R agonist measurements with explicit structures and potency $\leq$ 15,000 nM. Rows were excluded when the assay context was inconsistent with direct agonism, the structure was invalid, or the compound fell outside the screen's Stage-0 design space. Remaining compounds were deduplicated by canonical SMILES and scaffold-deduplicated to yield the in-domain active set.

Each active was paired with 30 unique decoys from the local ZINC pool. Decoys were required to match formal charge, minimize nearest-neighbor distance in molecular weight, LogP, HBA, HBD, topological polar surface area, and rotatable-bond count, differ in Murcko scaffold, and satisfy ECFP4 Tanimoto similarity < 0.35 to every active. Benchmark-mode screening then ranked the entire 310-molecule universe while preserving stage failures at the bottom of the ranked list under explicit status labels.

Ten in-domain actives is a small set, and it is the binding constraint on the GLP-1R benchmark's statistical power. It reflects how few nonpeptidic direct GLP-1R agonists with public structures fall inside a drug-like design space, rather than a restrictive filter on our part: Table S5 shows that 1,346 of the 1,892 excluded rows fell outside the Stage-0 design space and a further 432 came from assays that do not support a direct-agonist interpretation.

Table S4: In-domain GLP-1R agonists used in the retrospective benchmark. Potency values report the best retained ChEMBL measurement after canonical-SMILES deduplication, direct-agonist assay filtering, Stage-0 design-space filtering, and scaffold deduplication. Chemotype labels identify the unique scaffolds retained for active-recovery analyses.

Ligand ID	Source	Potency type	Potency (nM)	Chemotype
CHEMBL398714	ChEMBL:CHEMBL1147050	EC50	101	scaffold_1
CHEMBL6027640	ChEMBL:CHEMBL5726519	EC50	180	scaffold_2
CHEMBL250112	ChEMBL:CHEMBL1140467	EC50	433	scaffold_3
CHEMBL399841	ChEMBL:CHEMBL1140467	EC50	1,000	scaffold_4
CHEMBL251760	ChEMBL:CHEMBL1140467	EC50	1,200	scaffold_5
CHEMBL400700	ChEMBL:CHEMBL1140467	EC50	1,400	scaffold_6
CHEMBL401183	ChEMBL:CHEMBL1140467	EC50	2,200	scaffold_7
CHEMBL250748	ChEMBL:CHEMBL1140467	EC50	3,000	scaffold_8
CHEMBL248912	ChEMBL:CHEMBL1140467	EC50	5,100	scaffold_9
CHEMBL400500	ChEMBL:CHEMBL1140467	EC50	14,000	scaffold_10

Table S5: Summary of excluded ChEMBL activity rows during benchmark curation. Counts summarize why ChEMBL GLP-1R records were removed before active-set assembly. The dominant categories distinguish compounds outside the Stage-0 physicochemical design space from records whose assay context did not support a direct-agonist interpretation.

Exclusion reason	Count
Out of Stage-0 design space	1,346
Not a direct-agonist assay	432
Excluded assay context	71
Scaffold deduplicated	24
Invalid SMILES	16
Failed potency threshold	3
Total excluded	**1,892**

Additional Peptide-Receptor and Protein-Protein Benchmarks

To test generalization beyond GLP-1R, the workflow was applied to three additional interfaces using a fully automated procedure with no manual curation: two peptide-GPCRs (the ghrelin receptor, GHSR, and neurotensin receptor 1, NTSR1) and one protein-protein interaction interface (the p53-binding site of MDM2). For each system, 50 distinct-scaffold in-domain actives were curated from ChEMBL with the same Stage-0 design-space, canonical-SMILES, and Murcko-scaffold deduplication filters used for GLP-1R, and 30 decoys per active (1,500 total) were matched from the same local ZINC pool. The receptor-side pharmacophore and the native peptide-contact reference were generated automatically from the corresponding peptide-bound complex (PDB 7F9Y, 4GRV, and 1YCR).

The three additional interfaces broaden the evaluation beyond GLP-1R. Two are peptide-GPCRs and one is a protein-protein interface. One non-GPCR system cannot establish a general relationship between interface geometry and screening performance.

Table S6: Composition of the additional retrospective benchmarks. Each labeled universe contains 50 in-domain actives (distinct Murcko scaffolds) and 30 property-matched, scaffold-distinct, ECFP4-dissimilar decoys per active (1,500 total) drawn from the local ZINC pool.

System	ChEMBL target	Reference complex	Actives	Decoys
Ghrelin receptor (GHSR)	CHEMBL4616	PDB 7F9Y	50	1,500
Neurotensin receptor 1 (NTSR1)	CHEMBL4123	PDB 4GRV	50	1,500
MDM2–p53 interface	CHEMBL5023	PDB 1YCR	50	1,500

Table S7: In-domain actives used in the additional retrospective benchmarks. Actives were curated from ChEMBL with the same Stage-0 design-space, canonical-SMILES, and Murcko-scaffold deduplication filters used for GLP-1R; peptide-GPCR systems use functional agonist potencies and the MDM2–p53 interface uses p53-pocket binding/inhibition potencies. Each active has a distinct Murcko scaffold.

System	Ligand (ChEMBL)	Potency type	Potency (nM)	MW	cLogP
Ghrelin receptor (GHSR)	CHEMBL480821	EC50	0.0501	436	2.50
	CHEMBL4172656	EC50	0.1	431	3.13
	CHEMBL251572	EC50	0.159	485	3.76
	CHEMBL401134	EC50	0.2	479	3.70
	CHEMBL518774	EC50	0.2	437	1.08
	CHEMBL259166	EC50	0.2	494	0.41
	CHEMBL460680	EC50	0.27	498	1.64
	CHEMBL270666	EC50	0.3	480	2.02
	CHEMBL482220	EC50	0.316	482	3.76
	CHEMBL2364468	EC50	0.316	490	4.90
	CHEMBL260819	EC50	0.398	500	4.50
	CHEMBL271876	EC50	0.4	465	1.71
	CHEMBL404336	EC50	0.5	460	2.95
	CHEMBL251169	EC50	0.501	452	3.79
	CHEMBL2364429	EC50	0.501	490	4.86
	CHEMBL1161835	EC50	0.6	475	2.93
	CHEMBL2364354	EC50	0.631	473	4.11
	CHEMBL410135	EC50	0.64	496	1.24
	CHEMBL379074	EC50	0.794	492	4.92
	CHEMBL4637980	EC50	0.794	498	1.57
	CHEMBL2364408	EC50	0.794	478	4.78
	CHEMBL403180	EC50	0.8	462	2.87
	CHEMBL2364382	EC50	1	458	4.41
	CHEMBL2364338	EC50	1	445	3.78
	CHEMBL1923504	EC50	1	495	2.07
	CHEMBL4645148	EC50	1	499	0.97
	CHEMBL1161828	EC50	1.1	498	2.58
	CHEMBL516602	EC50	1.26	419	0.78
	CHEMBL4170481	EC50	1.26	433	2.32
	CHEMBL3218890	EC50	1.3	473	4.08
	CHEMBL4164328	EC50	1.58	390	2.28
	CHEMBL2364381	EC50	1.58	464	4.47

System	Ligand (ChEMBL)	Potency type	Potency (nM)	MW	cLogP
	CHEMBL2364384	EC50	1.58	460	4.49
	CHEMBL2364480	EC50	2	490	4.86
	CHEMBL481620	EC50	2	487	2.05
	CHEMBL4637284	EC50	2	500	0.36
	CHEMBL519322	EC50	2	387	4.20
	CHEMBL1161827	EC50	2	482	2.91
	CHEMBL251573	EC50	2.51	475	3.35
	CHEMBL272540	EC50	2.51	449	3.40
	CHEMBL480639	EC50	2.51	464	3.15
	CHEMBL4164040	EC50	2.51	447	3.25
	CHEMBL1161829	EC50	2.69	450	3.18
	CHEMBL1162058	EC50	3	486	4.85
	CHEMBL130229	EC50	3	479	2.61
	CHEMBL382844	EC50	3	492	1.37
	CHEMBL4171914	EC50	3	443	3.12
	CHEMBL2364485	EC50	3.16	457	3.65
	CHEMBL1079185	EC50	3.98	439	4.15
	CHEMBL1161822	EC50	4.79	466	3.90
Neurotensin receptor 1 (NTSR1)	CHEMBL463811	EC50	178	495	3.61
	CHEMBL294989	EC50	216	353	4.02
	CHEMBL462830	EC50	220	493	4.98
	CHEMBL2431120	EC50	2e+03	421	3.86
	CHEMBL3315193	EC50	2.05e+03	415	4.47
	CHEMBL3099769	EC50	2.3e+03	488	4.94
	CHEMBL3315203	EC50	3.24e+03	415	4.47
	CHEMBL1301125	EC50	3.27e+03	488	4.50
	CHEMBL3315221	EC50	3.28e+03	415	4.47
	CHEMBL1323442	EC50	3.35e+03	393	1.76
	CHEMBL1490404	EC50	3.8e+03	364	3.36
	CHEMBL3099777	EC50	3.8e+03	489	4.33
	CHEMBL1611170	EC50	3.89e+03	353	1.82
	CHEMBL1600146	EC50	3.91e+03	303	1.45
	CHEMBL1468368	EC50	4.12e+03	419	2.20
	CHEMBL3315213	EC50	4.33e+03	450	4.67
	CHEMBL3315202	EC50	4.96e+03	415	4.47
	CHEMBL1546134	EC50	5.14e+03	334	3.49
	CHEMBL3315209	EC50	5.17e+03	440	4.27
	CHEMBL1538066	EC50	5.4e+03	463	3.79
	CHEMBL2431121	EC50	5.8e+03	449	4.64
	CHEMBL1597454	EC50	5.83e+03	402	3.96
	CHEMBL1446377	EC50	5.9e+03	435	4.25
	CHEMBL1547420	EC50	5.98e+03	401	2.91
	CHEMBL1312502	EC50	6e+03	450	4.89
	CHEMBL1328169	EC50	6.21e+03	437	4.95
	CHEMBL1586252	EC50	6.24e+03	362	4.35
	CHEMBL1594369	EC50	6.59e+03	370	3.82
	CHEMBL2003819	EC50	6.62e+03	403	3.21
	CHEMBL3315194	EC50	6.65e+03	288	2.26
	CHEMBL1550591	EC50	6.66e+03	273	3.56
	CHEMBL2431119	EC50	7e+03	406	3.63
	CHEMBL1966869	EC50	7.15e+03	497	1.54
	CHEMBL1438069	EC50	7.37e+03	335	3.81
	CHEMBL3315223	EC50	7.46e+03	415	4.47
	CHEMBL1611686	EC50	7.49e+03	498	2.52
	CHEMBL1595121	EC50	7.5e+03	322	4.57
	CHEMBL1453900	EC50	7.59e+03	369	4.13
	CHEMBL1340713	EC50	7.59e+03	463	0.90
	CHEMBL3315204	EC50	7.72e+03	431	4.55
	CHEMBL1461496	EC50	7.86e+03	374	3.53
	CHEMBL1387563	EC50	7.89e+03	374	4.51
	CHEMBL1321527	EC50	8.09e+03	221	2.50
	CHEMBL1385214	EC50	8.21e+03	309	3.41
	CHEMBL1369296	EC50	8.29e+03	300	3.39
	CHEMBL1995726	EC50	8.51e+03	376	3.10
	CHEMBL1608453	EC50	8.64e+03	350	3.75
	CHEMBL1586169	EC50	8.85e+03	485	2.36

System	Ligand (ChEMBL)	Potency type	Potency (nM)	MW	cLogP
	CHEMBL1493442	EC50	8.91e+03	413	3.29
	CHEMBL1325640	EC50	9.06e+03	257	2.79
MDM2–p53 interface	CHEMBL3407579	Ki	0.13	496	4.55
	CHEMBL5913446	IC50	1	494	4.66
	CHEMBL3653239	IC50	1.47	493	4.96
	CHEMBL5989834	IC50	3.66	484	4.29
	CHEMBL3693969	Ki	4.3	476	4.53
	CHEMBL5879668	IC50	4.54	496	4.62
	CHEMBL5883242	IC50	5.34	495	4.44
	CHEMBL5921912	IC50	6.28	495	4.72
	CHEMBL378100	Ki	8.5	474	4.83
	CHEMBL4102803	IC50	9.6	389	3.73
	CHEMBL5869704	IC50	9.75	496	4.43
	CHEMBL5978015	IC50	13.1	471	4.18
	CHEMBL3819071	IC50	15.2	492	3.86
	CHEMBL3687241	IC50	16.2	486	4.76
	CHEMBL6064615	IC50	16.4	478	3.78
	CHEMBL5849977	IC50	18	452	4.47
	CHEMBL5876084	IC50	25	466	4.48
	CHEMBL6024824	IC50	25.4	495	4.59
	CHEMBL5877121	IC50	26	437	2.38
	CHEMBL5859878	IC50	29.8	498	4.11
	CHEMBL5267044	IC50	30	402	4.50
	CHEMBL5816136	IC50	30	486	4.85
	CHEMBL5988926	IC50	35.8	493	3.98
	CHEMBL6024197	IC50	36	439	4.62
	CHEMBL5901726	IC50	36.2	497	3.17
	CHEMBL4115504	IC50	40	462	4.91
	CHEMBL199341	IC50	50	214	2.30
	CHEMBL4114636	IC50	50	492	4.42
	CHEMBL5811937	IC50	62.8	500	3.91
	CHEMBL6030067	IC50	69.8	497	4.65
	CHEMBL6006846	IC50	71	498	4.11
	CHEMBL5854116	IC50	88	399	4.98
	CHEMBL2177185	IC50	90	456	4.83
	CHEMBL4109769	IC50	90	473	4.66
	CHEMBL2313485	IC50	90	378	4.27
	CHEMBL514709	Ki	93	281	3.20
	CHEMBL5808134	IC50	95	425	4.32
	CHEMBL5745234	IC50	98.8	455	4.56
	CHEMBL2347389	IC50	100	434	4.89
	CHEMBL5775440	IC50	101	386	4.26
	CHEMBL5747471	IC50	132	424	4.92
	CHEMBL3265101	Ki	140	370	4.87
	CHEMBL4068667	IC50	140	462	4.27
	CHEMBL5928219	IC50	143	471	3.66
	CHEMBL3220103	IC50	150	478	4.47
	CHEMBL6012158	IC50	161	499	4.59
	CHEMBL1688303	IC50	170	479	4.89
	CHEMBL3217780	IC50	170	495	4.63
	CHEMBL416288	IC50	175	267	2.85
	CHEMBL6050672	IC50	189	499	4.59

Table S8: Robustness of the full cascade across five independent matched-decoy sets. Values are mean ± sample standard deviation over five runs per system. Each replicate contains the 10-active set and 300 matched decoys, with structural alerts recorded rather than excluding; these replicates are distinct from the 50-active comparisons.

System	ROC-AUC	PR-AUC	BEDROC
GLP-1R	0.735 ± 0.008	0.366 ± 0.066	0.477 ± 0.020
GHSR	0.750 ± 0.000	0.516 ± 0.000	0.605 ± 0.000
NTSR1	0.899 ± 0.001	0.782 ± 0.017	0.849 ± 0.005
MDM2–p53	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000

Robustness Across Decoy Sets

Five independently sampled matched-decoy sets per system were scored from molecular inputs with structural alerts recorded rather than excluding. Each contains 10 actives and 300 decoys; these replicate sets are distinct from the 50-active benchmarks. Table S8 reports the measured mean and sample standard deviation. All decoys share the same ZINC source, so repeated draws cannot exclude systematic source-related bias.

Table S9: Redocking of the top-10 final-ranked GLP-1R ligands. Affinities and active-minus-inactive differences are in kcal/mol. Each state value is the best successful result across its receptor structures and three seeds. Missing paired results are shown as dashes and excluded from the paired test; all requested jobs remain in the raw output.

Rank	Ligand ID	Active	Inactive	Δ
1	ZINCk700000Gfz1H	-7.18	-5.71	-1.47
2	ZINCk700001bQgxo	-7.34	-6.22	-1.12
3	ZINCk500000BlHKL	-7.07	-5.88	-1.19
4	ZINCj600000JWMq0	-8.45	-7.07	-1.39
5	ZINCk500000gdMQW	-7.05	-6.13	-0.92
6	ZINCk600000EmAGZ	-8.56	-7.25	-1.31
7	ZINCk5000003XjkE	-7.21	-6.12	-1.08
8	ZINCk800000gvPBH	-6.83	-6.40	-0.44
9	ZINCk600000Ja5bY	-8.70	-7.24	-1.46
10	ZINCk600000y8tEo	-7.42	-6.28	-1.14

Orthogonal Redocking of the Top-Ranked Ligands

The new top 10 were prepared and redocked against three active-state (6X18, 7KI0, 7LCJ) and two inactive-state (5VEW, 6LN2) GLP-1R structures, with three seeds per structure and exhaustiveness 16. Of 10 selected ligands, 10 have successful results against both states. The median active-minus-inactive difference is -1.16 kcal/mol; 10 paired ligands favor the active state. An exact signed-rank test using exhaustive sign flips, average ranks for ties, and zero differences excluded gives $W = 0$ and two-sided $p = 0.0020$. This is a computational score comparison, not evidence of agonism. The raw output accounts for every requested job, including preparation and docking failures.

Where Actives Are Lost Under Production Constraints

Each system's actives and matched decoys were screened inside 30,000 background molecules, with structural alerts recorded rather than excluding and the property limits retained. Table S10 identifies every active absent from the final native ranking, including losses after shortlist admission. Table S11 gives measured stage survival under the 5% rule with a 1,000-molecule minimum. All 34 alert-flagged actives pass Stage 0; only two NTSR1 actives fail the property envelope. Later losses remain attributable to the hotspot gate, shortlist, preparation, native selection, or final ranking as recorded per molecule.

Table S10: In-domain actives absent from the final ranking under production constraints. Structural alerts are annotations, not exclusions. Causes refer to the first failed or limiting stage.

System	Ligand	Stage	Cause
GLP-1R	CHEMBL398714	Stage 1	required hotspot matches or contact groups not met
GLP-1R	CHEMBL401183	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
GHSR	CHEMBL480821	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL4172656	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL518774	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL460680	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL482220	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL1161828	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL516602	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL519322	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL480639	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL1161829	Stage 1	required hotspot matches or contact groups not met
GHSR	CHEMBL1161822	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL294989	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1323442	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1600146	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1586252	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL3315194	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1550591	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1438069	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1611686	Stage 0	property: logp=7.27
NTSR1	CHEMBL1595121	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1340713	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1321527	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1385214	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1369296	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1608453	Stage 1	required hotspot matches or contact groups not met
NTSR1	CHEMBL1586169	Stage 0	property: logp=5.36
MDM2–p53	CHEMBL378100	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL5267044	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL199341	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL2177185	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL514709	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL2347389	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates
MDM2–p53	CHEMBL416288	Shortlist	outside the 5% shortlist with a minimum of 1,000 eligible candidates

Table S11: Stage-by-stage active survival under production constraints. Each labeled library sits in a 30,000-molecule background, with structural alerts recorded rather than excluding and a minimum of 1,000 molecules on the 5% shortlist. Entries count actives and include the measured native-preparation and scoring outcomes.

Stage	GLP-1R	GHSR	NTSR1	MDM2–p53
Input	10	50	50	50
Stage 0	10	50	48	50
Stage 1	9	39	35	50
Shortlist	8	39	35	43
Stage 3	8	39	35	43
Native selection	8	39	35	43
Native scoring	8	39	35	43
Final top-1,000	8	39	35	43

Million-Compound Screen and Paired Shortlist Rules

The million-compound screen ran over all 1,000,000 inputs with structural alerts recorded rather than excluding and a capped 1,000-molecule minimum on its 5% shortlist. Its percentage denominator is the Stage-0-pass count; Table S1 gives its full configuration and stage counts.

Paired production runs isolate the shortlist minimum while holding the alert policy fixed. These pairs use the production-benchmark denominator, namely candidates surviving Stages 1/2. Each pair shares one evaluation of prescreening and the larger Stage-3 union, with separate native preparation and scoring when the shortlists differ; identical shortlists explicitly share a single native execution. Tables S12 and S13 report complete counts and measured native-branch wall times, respectively.

Table S12: Stage-by-stage counts for each pair of shortlist rules. Entries are total ligands and in-domain actives, respectively. All systems use the same 30,000-molecule background, fixed 0.25/0.75 production-benchmark weights, and the candidate count as the percentage denominator.

		5% only		5% + min. 1,000	
System	Stage	Total	Actives	Total	Actives
GLP-1R	Input	30,310	10	30,310	10
GLP-1R	Stage 0	30,126	10	30,126	10
GLP-1R	Stages 1/2	29,274	9	29,274	9
GLP-1R	Shortlist	1,464	8	1,464	8
GLP-1R	Successful 3D	1,462	8	1,462	8
GLP-1R	Native selection	647	8	647	8
GLP-1R	Successful native	647	8	647	8
GLP-1R	Final ranking	647	8	647	8
GHSR	Input	31,550	50	31,550	50
GHSR	Stage 0	31,366	50	31,366	50
GHSR	Stages 1/2	277	39	277	39
GHSR	Shortlist	14	9	277	39
GHSR	Successful 3D	14	9	277	39
GHSR	Native selection	14	9	271	39
GHSR	Successful native	14	9	271	39
GHSR	Final ranking	14	9	271	39
NTSR1	Input	31,550	50	31,550	50
NTSR1	Stage 0	31,364	48	31,364	48
NTSR1	Stages 1/2	398	35	398	35
NTSR1	Shortlist	20	18	398	35
NTSR1	Successful 3D	20	18	397	35
NTSR1	Native selection	20	18	343	35
NTSR1	Successful native	20	18	343	35
NTSR1	Final ranking	20	18	343	35
MDM2–p53	Input	31,550	50	31,550	50
MDM2–p53	Stage 0	31,366	50	31,366	50
MDM2–p53	Stages 1/2	31,314	50	31,314	50
MDM2–p53	Shortlist	1,566	43	1,566	43
MDM2–p53	Successful 3D	1,566	43	1,566	43
MDM2–p53	Native selection	1,152	43	1,152	43
MDM2–p53	Successful native	1,152	43	1,152	43
MDM2–p53	Final ranking	1,000	43	1,000	43

Table S13: Measured native-branch wall times for each pair of shortlist rules. Values cover native selection, ligand preparation, scoring, and output generation. Stages 0–2 and the larger Stage-3 union are shared within each pair and are excluded here. Identical shortlists share one native execution. These runs execute alongside the million-compound screen, so the elapsed times are workload-specific measurements rather than portable speedup estimates.

System	5% only (min)	5% + min. 1,000 (min)	Native execution
GLP-1R	17.59	17.59	Shared (identical input)
GHSR	2.35	8.83	Separate per policy
NTSR1	3.26	8.88	Separate per policy
MDM2–p53	41.66	41.66	Shared (identical input)

Table S14: Sensitivity of benchmark enrichment to how equal scores are ordered. The ID columns order molecules with equal scores by ligand identifier; the tied columns, used throughout this work, leave them tied and average over the possible within-tie orders. Molecules with equal status and score are tied, including unscored molecules within a failure status. Both columns evaluate the same molecular scores on the same molecules.

System	Method	ROC ID	ROC tied	PR ID	PR tied
GLP-1R	Full cascade	0.796	0.753	0.464	0.363
	Native-only	0.809	0.758	0.463	0.361
	Stage-3 only	0.738	0.727	0.285	0.284
	Single-pass 3D	0.717	0.717	0.259	0.259
GHSR	Full cascade	0.965	0.867	0.732	0.610
	Native-only	0.952	0.926	0.819	0.680
	Stage-3 only	0.947	0.852	0.440	0.408
	Single-pass 3D	0.794	0.794	0.231	0.231
NTSR1	Full cascade	0.899	0.785	0.523	0.390
	Native-only	0.744	0.690	0.385	0.127
	Stage-3 only	0.884	0.778	0.388	0.362
	Single-pass 3D	0.560	0.560	0.142	0.142
MDM2–p53	Full cascade	0.960	0.945	0.910	0.705
	Native-only	0.960	0.945	0.910	0.705
	Stage-3 only	0.339	0.339	0.023	0.023
	Single-pass 3D	0.339	0.339	0.023	0.023

Tie-Aware Benchmark Evaluation

Ligand identifiers encode different naming conventions for ChEMBL actives and ZINC decoys, so ordering equal scores by identifier lets a naming artifact inflate rank-based enrichment. Every benchmark statistic reported here therefore retains equal-status, equal-score ties, including missing scores within a failure status. ROC-AUC and average precision operate at tied score thresholds. EF, BEDROC, and top-k retrieval average over every possible within-tie ordering, so expected active counts can be fractional. EF is normalized by the actual selected fraction $\lceil fN \rceil / N$ rather than the unrounded nominal fraction. Grouped-bootstrap intervals and paired comparisons use the same tie treatment and EF normalization. How equal scores are ordered affects only the evaluation: the molecular scores, production shortlists, and observed final survival counts do not depend on it.

Table S14 quantifies how much the estimates depend on that choice, setting an identifier-ordered calculation beside the tie-aware ROC-AUC and PR-AUC used throughout. Machine-readable records provide both the emitted ranks and the evaluation tie groups; the independent reproduction script computes the reported statistics from those groups. Validation checks cover exhaustive within-tie permutations, scikit-learn ROC-AUC and average precision, identifier invariance, and explicit expansion of bootstrap samples.